The cylinder's cross sections are circles in the diagram, but if we were to restore the third dimension of our brane by moving out of Gargantua's equatorial plane, the cross sections would become flattened spheres (spheroids).

On Gargantua's equatorial plane I marked several special locations that occur in my science interpretations of *Interstellar*: Gargantua's event horizon (black circle), the critical orbit from which Cooper and TARS fall into Gargantua near the end of the movie (green circle; Chapter 27), the orbit of Miller's planet (blue circle; Chapter 17), the orbit in which the *Endurance* is parked while the crew visit Miller's planet (yellow circle), and a segment of the nonequatorial orbit of Mann's planet, projected into the equatorial plane (purple circle). The outer part of Mann's orbit is so far away from Gargantua (600 Gargantua radii or more; Chapter 19) that I had to redraw the picture on a much larger scale to fit it in (bottom picture), and, even so, I didn't do it honestly: I only put the outer part at 100 Gargantua radii instead of 600 as I should. The red circles are labeled "SOF" for "shell of fire"; see below.

How did I come up with these locations? I use the parking orbit as an illustration here and discuss the others later. In the movie, Cooper describes the parking orbit this way: "So we track a wider orbit of Gargantua, parallel with Miller's planet but a little further out." And he wants it to be far enough from Gargantua to be "out of the time shift," that is, far enough from Gargantua that the slowing of time compared to Earth is very modest. This motivated my choice of five Gargantua radii (yellow circle in Figure 6.3). The time for the Ranger to travel from this parking orbit to Miller's planet, two and a half hours, reinforced my choice.

But there was a problem with this choice. At this distance, Gargantua would look huge; it would subtend about 50 degrees on the